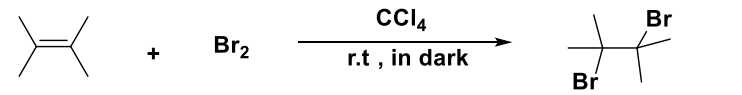
Class 1

● Bromination Reaction



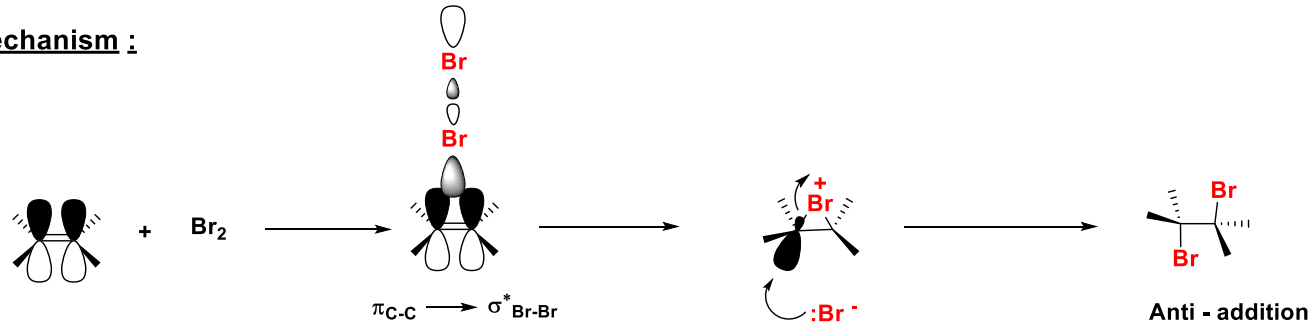
An alkene colorless

Brown

Colorless Vicinal dibromide

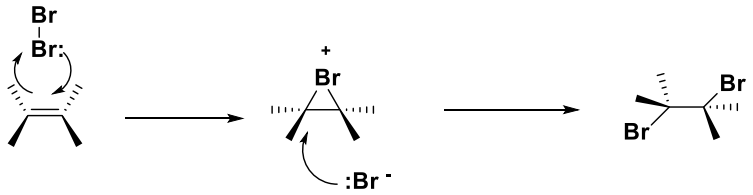
Rapid decolorization of Br_2/CCl_4 (Classical test for alkene and alkynes)

Mechanism :



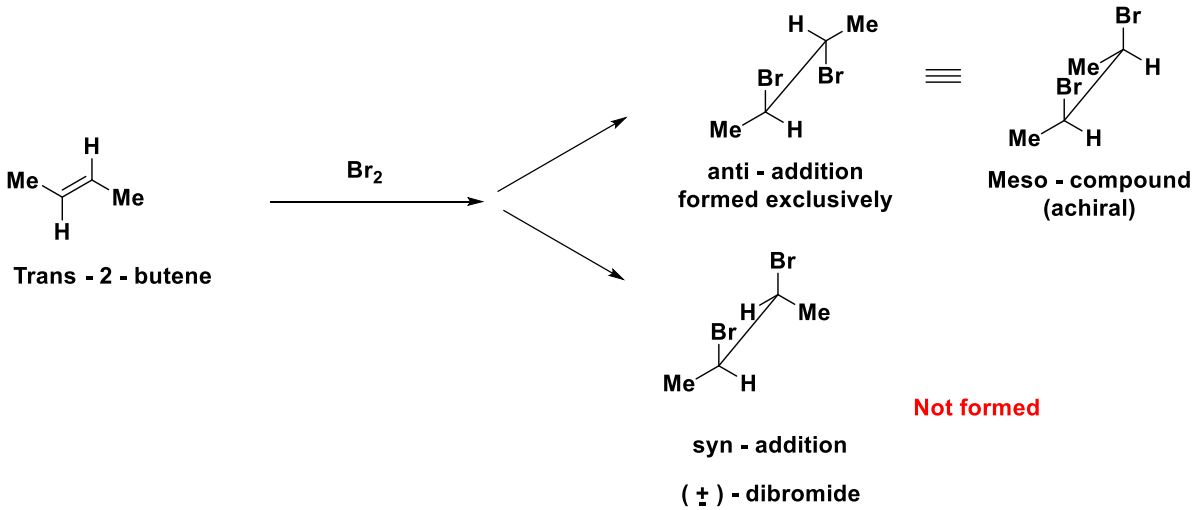
Formation of a π – complex
or
charge transfer complex

For arrow pushing



● Evidence in supporting this mechanism

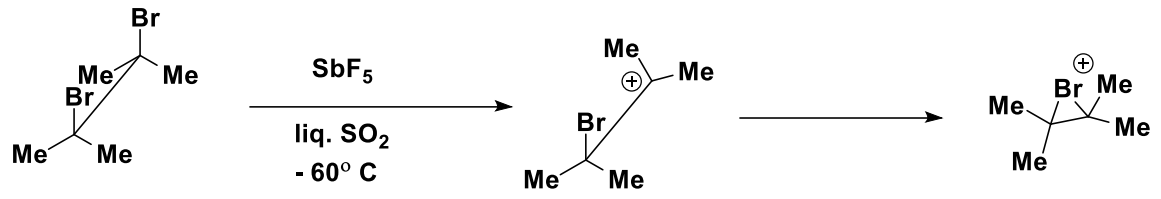
1.



It can only be explained by the formtion of a π – complex and cyclic bromonium ion intermediate, followed by the attack of Br^- from the opposite side.

Lewis acid such as $AlBr_3$ accelerate the rate of the reaction $\Longrightarrow$ formation of bromonium ion is the r.d.s

2.

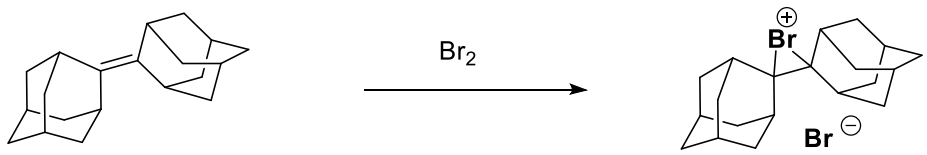


δ 2.0 **ppm**
symmetric molecule - all CH3 groups are equal

expected two signals for CH_3 not observed

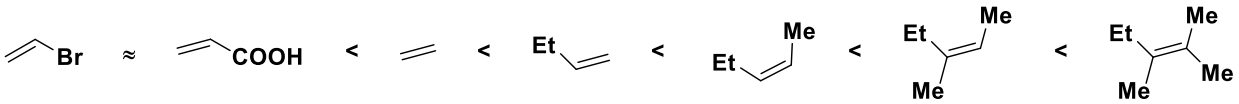
δ **2.9 ppm**
one signal is observed indicative of symmetric molecule

3.



isolated and characterised. stable compound as back side attack is not possible here.

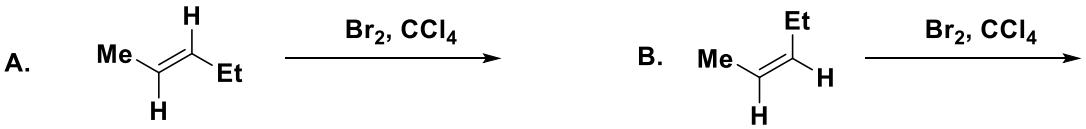
- **Effects of substituents on the rate of reaction**



	0.03		1	96	4160	11900	92500
		≈	1	100	4000	10000	100000
		≈	1	10^2	$4*10^3$	10^4	10^6

Questions:

1. The addition rate of bromine to ethylene increases by adding AlBr3, but decreases by addition of KBr. Explain.
2. Addition of bromine to alkene proceeds through a twostep pathway and not through a four centered one step (concerted) pathway. Give experimental evidences to account for the fact.
3. On bromination cis-2-butene gives only dl-2,3-dibromobutane and trans-2-butene gives only the meso-dibromide product. Give the mechanism.
4. Give (R) – (S) designation of the products that you would expect from each of the following reactions.



5. Predict the major product when butadiene is treated with bromine in methanol as solvent.
6. Define stereoselective and stereospecific reaction. Give examples.